Data Structures and Algorithms (3 Credits)

Course Outline:

Module 1: Algorithm Analysis (time and space complexity, Big-O notation, analysis examples).
Module 2: Introduction to Data Structures (arrays and tensor representations, linked lists, stacks, queues, applications).
Module 3: Trees (binary trees and their traversals, general tree representations, binary search trees and operations on them, Huffman codes, heaps and operations on them, dynamic memory management, garbage collection, applications).
Module 4: Sorting and Searching Algorithms (quicksort, mergesort, binary search).
Module 5: Advanced Data Structures (hash tables, tries and string matching, graphs and their representations, DFS, BFS, minimum spanning trees, shortest path algorithms, topological sorting, examples of balanced trees, applications).

Note: The topics in the modules above may be taught in a different order than indicated above so as to enhance the understanding. Examples of data structures and algorithms will also be drawn from AI applications (wherever possible). The students should note that fundamentals of DSA are essential to all topics of CS and AI. LLVMs generate code with many of these data structures in them.

References and Text books

T.H. Cormen, C.E. Leiserson, and R.L. Rivest, Introduction to Algorithms, The MIT Press, Cambridge, Massachusetts, USA, 1990.

M.A. Weiss, Data Structures and Algorithms Analysis in C++, Benjamin/Cummins, Redwood City, California, USA, 1994.

M.T Goodrich, R. Tamassia, and M.H. Goldwasser, Data Structures and Algorithms in Python, Wiley, 2013.

I do not recommend any single text book. Students may read the material from any book that they are comfortable with.